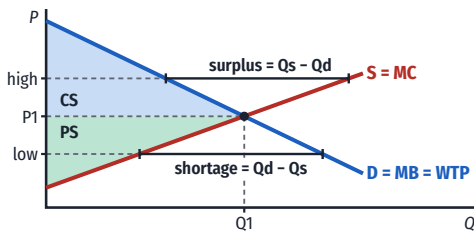


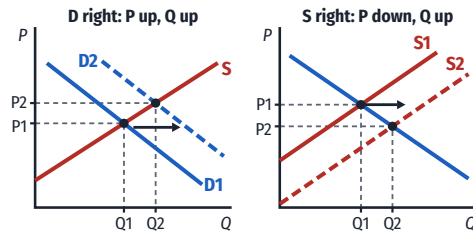
1 Market equilibrium and surplus



Label: $D = MB = WTP$, $S = MC$, P_1 , Q_1 , CS, PS; Q_s = amount sellers offer, Q_d = amount buyers want at a given price
 1. Draw D and S ; mark where they cross: P_1 , Q_1 .
 2. Shade CS (under D , above P_1) and PS (above S , below P_1).

- $CS = \frac{1}{2} \times Q_1 \times (\text{where } D \text{ hits the } P \text{ axis} - P_1)$.
- $PS = \frac{1}{2} \times Q_1 \times (P_1 - \text{where } S \text{ hits the } P \text{ axis})$.
- Total surplus $CS + PS$ is largest at Q_1 , so there is no DWL.
- Above P_1 : $Q_s > Q_d$ (surplus), P falls. Below: shortage, P rises.

2 Shifting demand and supply

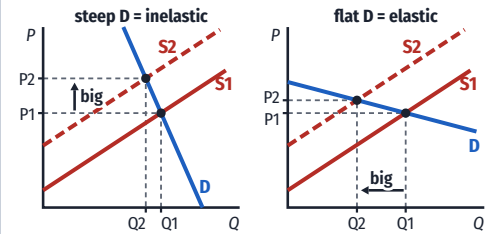


Label: D_1 to D_2 (or S_1 to S_2) with an arrow; P_1 , Q_1 , P_2 , Q_2

1. Buyer changes (income, tastes) shift D ; cost changes shift S .
2. Draw the shift with an arrow; mark the new crossing P_2 , Q_2 .

- D right (more demand): P up, Q up. D left: P down, Q down.
- S right (more supply): P down, Q up. S left: P up, Q down.
- Both shift: D right and S left give P up; both right give Q up; the other one depends on which shift is bigger.
- The good's own price moves along a curve; it never shifts it.

3 Elasticity: steep versus flat demand

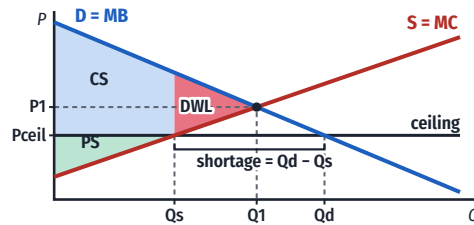


Label: steep D = inelastic, flat D = elastic; S_1 to S_2 ; P_1 , Q_1 , P_2 , Q_2

1. Draw the same S_1 to S_2 shift on both graphs.
2. Steep D : P changes a lot, Q a little. Flat D : the reverse.

- ϵ (elasticity) = % change in $Q \div$ % change in P : how strongly buyers react. Negative for D ; $|\epsilon|$ = its size.
- On a straight D : $\epsilon = (1 \div \text{slope of } D) \times (P \div Q)$ at that point.
- $|\epsilon| > 1$ elastic, $|\epsilon| < 1$ inelastic. Vertical D : 0. Flat D : infinite.
- Spending $P \times Q$ rises with P if inelastic, falls if elastic.

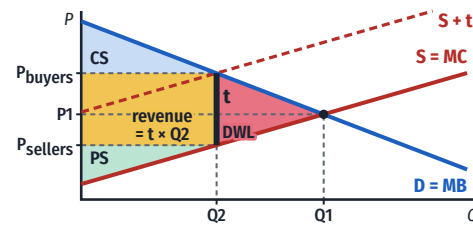
4 Price ceiling (and floor)



Label: P_{ceil} = legal maximum price; at P_{ceil} , Q_s = amount sellers offer (on S), Q_d = amount buyers want (on D)
 1. Draw D and S crossing at P_1 , Q_1 . Neither curve shifts.
 2. Flat P_{ceil} line below P_1 . Shade DWL from Q_s to Q_d .

- Matters only below P_1 . Units sold = Q_s . Shortage = $Q_d - Q_s$.
- $DWL = \frac{1}{2} \times (Q_1 - Q_s) \times (\text{height of } D \text{ at } Q_s - P_{ceil})$.
- PS falls. CS can rise (lower price) or fall (fewer units).
- Floor (legal minimum) above P_1 : sold = Q_d , surplus = $Q_s - Q_d$.

5 Per-unit tax

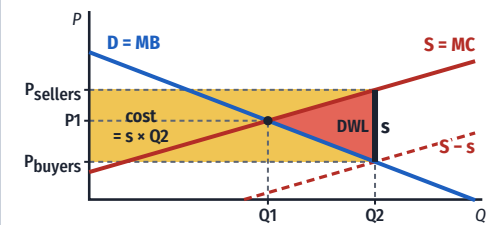


Label: $D = MB$, $S = MC$, $S + t$, P_1 , Q_1 , Q_2 , P_{buyers} , $P_{sellers}$; t = tax per unit, P_{buyers} = price buyers pay, $P_{sellers}$ = price sellers keep

1. Draw D and S at P_1 , Q_1 , then $S + t$: S moved up by t .
2. Q_2 : $S + t$ meets D . Read P_{buyers} on D ; $P_{sellers} = P_{buyers} - t$ on S .
3. Shade revenue box (0 to Q_2), DWL from Q_2 to Q_1 , CS, PS.

- Tax revenue = $t \times Q_2$. $DWL = \frac{1}{2} \times t \times (Q_1 - Q_2)$.
- Steeper side pays more. Same if buyers pay legally ($D - t$).
- Burden per unit: buyers $P_{buyers} - P_1$, sellers $P_1 - P_{sellers}$.

6 Subsidy

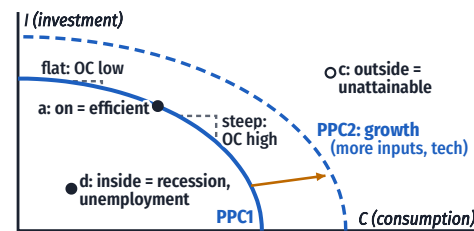


Label: $D = MB$, $S = MC$, $S - s$, P_1 , Q_1 , Q_2 , $P_{sellers}$, P_{buyers} ; s = subsidy per unit, $P_{sellers}$ = price sellers get, P_{buyers} = price buyers pay

1. Draw D and S at P_1 , Q_1 , then $S - s$: S moved down by s .
2. Q_2 : $S - s$ meets D ($Q_2 > Q_1$). P_{buyers} on D ; $P_{sellers} = P_{buyers} + s$.
3. Shade the cost box (0 to Q_2) and DWL from Q_1 to Q_2 .

- Government cost = $s \times Q_2$. $DWL = \frac{1}{2} \times s \times (Q_2 - Q_1)$.
- DWL because units from Q_1 to Q_2 have $MC > MB$.
- CS gain + PS gain = government cost - DWL .

7 PPC (production possibilities curve)

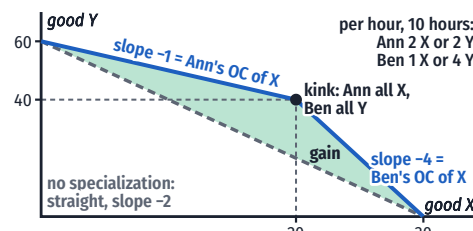


Label: axes C (consumption), I (investment); PPC_1 : points a (on), d (inside), c (outside); PPC_2 = growth

1. Bowed-out PPC_1 joining the two intercepts; mark a, d, c.
2. Growth (more workers, machines, tech): PPC_2 farther out.

- Slope = $-OC$ of one more C . Bowed out: that OC rises.
- Inside = unemployment or recession, not an inward shift. More I today = PPC farther out later.

8 Comparative advantage: joint PPC

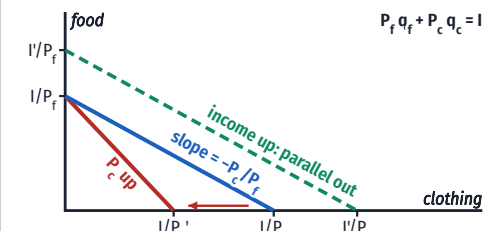


Label: intercepts, kink (where slope changes), slopes = OC s, gain

1. Intercepts = sum of everyone's max: 60 Y , 30 X in 10 hours.
2. From 60 Y , the person with lower OC of X goes first (Ann, slope -1) to the kink, then Ben (slope -4) to 30 X .

- OC of $X = Y$ per hour $\div X$ per hour (Ann 1, Ben 4). Lower OC of X = comparative advantage: make X .
- Three or more people: add pieces by OC of X , lowest first.

9 Budget line and spending rule

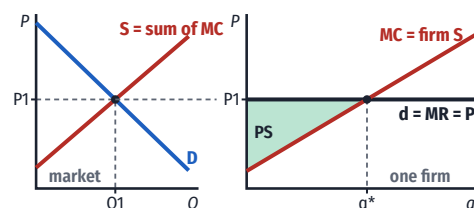


Label: food up, clothing across; I = income; P_f , P_c = prices; q_f , q_c = amounts bought; intercepts I/P_f , I/P_c ; slope $-P_c/P_f$

1. Join I : P_f on the food axis and $I \div P_c$ on clothing.
2. Income up (I'): parallel out. P_c up (P_c'): pivot in on clothing.

- Rational Spending Rule: $MU_a \div P_a = MU_b \div P_b$ (MU = extra satisfaction from one more unit; a, b = any two goods).
- P_c up: $MU_c \div P_c$ falls, so buy less clothing until equal again.

10 Competitive firm and market

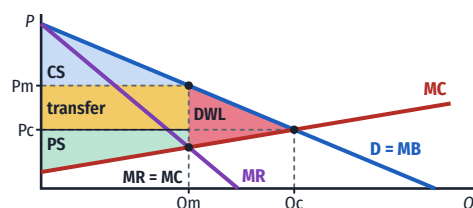


Label: market: D , S = sum of firms' MC , P_1 , Q_1 ; firm: flat $d = MR = P$ at P_1 (d = demand facing one firm), MC = firm supply, q^* , PS

1. Market: D and S cross at P_1 . Carry P_1 across: flat $d = MR = P$.
2. Firm: rising MC ; best output q^* where $MC = P$; shade PS .

- MR (marginal revenue) = extra revenue from one more unit; a firm too small to move the price gets $MR = P$.
- Produce where $P = MC$, so firm supply = its MC curve.
- Industry S : add all firms' q (one firm's quantity) at each P .
- Profit = PS - fixed cost. Firms enter if $PS >$ fixed cost.

11 Monopoly

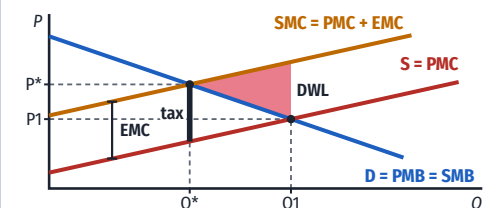


Label: $D = MB$; MR = marginal revenue; MC ; Q_m , P_m = monopoly Q and P ; Q_c , P_c = competitive Q and P ; CS, transfer, PS, DWL

1. D , then MR : same P intercept, twice as steep; rising MC .
2. Q_m where $MR = MC$; go straight up to D for P_m .
3. Q_c , P_c where D meets MC . DWL from Q_m to Q_c .

- D : $P = a - bQ$ (a = intercept, b = slope) gives $MR = a - 2bQ$.
- Transfer: CS in the box between P_c and P_m becomes profit.
- Profit = transfer + PS = revenue - area under MC .
- Given P_m , Q_m , flat MC : $a = 2 \times P_m - MC$, $b = (P_m - MC) \div Q_m$.

12 Negative externality and Pigouvian tax



Label: PMC = sellers' cost, EMC = harm to others, $SMC = PMC + EMC$; $D = PMB = SMB$ (private, social benefit); Q_1 , Q^* , DWL , tax

1. D and $S = PMC$ cross at Q_1 . Draw SMC , EMC above S .
2. Q^* (best amount, usually not 0) and P^* where SMC meets D . DWL from Q^* to Q_1 , pointing at Q^* .

- $DWL = \frac{1}{2} \times EMC \times (Q_1 - Q^*)$ if EMC is constant.
- Pigouvian tax = EMC at Q^* (per unit) moves the market to Q^* .
- Positive externality: $SMB = PMB + EMB$ (benefit to others) above D ; too little made; subsidy = EMB at Q^* .

Symbols: P price · Q quantity · 1 before, 2 after · D = MB (marginal benefit) = WTP (willingness to pay) · S = MC (marginal cost) · CS, PS consumer, producer surplus · DWL deadweight loss · OC opportunity cost · graph N = side 1 · other symbols: each panel's Terms line

How to answer (his grading)

- Terms: WTP = willingness to pay. MU = marginal utility.

If the question asks	Do this
Explain, why	answer, then reason (2 to 6 sentences)
Draw	label axes, curves; dotted lines, values
D down, S up: why	MB (MU, WTP) falls; MC rises as Q rises
What PS is	P received minus each unit's MC
Best pollution	social MC = social MB , usually above 0
Monopoly	MR (marginal revenue) = MC , P off D

D = MB = WTP. S = MC. Unclear: state your reading.

Opportunity cost and the PPC

graph 7

- Terms: OC = value of the next best option given up. Sunk = spent for good. PPC = most of two goods you can make.

If the question asks	Do this
a X or b Y: OC of 1 X	b ÷ a units of Y
Time, a ticket	hours × wage (about \$36) + resale value; price paid is sunk
Cost of college	tuition, fees, books + foregone earnings (about \$60,000); not rent, food
OC on a PPC	vertical lost ÷ horizontal gained
Bowed out	OC rises: best-suited switch first
Inside; shifts out	recession (no shift); more workers, capital, technology

PPC slope = minus the OC of the horizontal-axis good.

Comparative advantage

graph 8

- Terms: absolute advantage = more per hour. Comparative advantage = lower OC.

If the question asks	Do this
Who specializes	OC of 1 X = Y ÷ X made in the same time; lower OC makes X
All gain same %	OC order holds: no change
Equal OCs	no gains; same good, one PPC segment
Joint PPC	no specializing: straight; specializing: bowed out, lowest OC first (graph 8)
Trade price	between the two OCs
"Always", "solely"	usually false: history, policy matter

Lower OC decides, not absolute advantage.

Supply and demand

graphs 1, 2

- Terms: inferior good: bought less as income rises (normal: more). Substitute: used instead; complement: with it.

D right if (reverse: left)	S right if (reverse: left)
income up (normal good)	input prices fall
income down (inferior good)	better technology
substitute's P up	more sellers
complement's P down	tax or rule cut
more buyers; tastes for it	expect lower future P
expect higher future P	

If the question asks	Do this
Supply or demand?	P and Q same way: D. Opposite: S. Weather, disease: S.
"P rose, D fell"	wrong: a move along D
Ceiling, floor	neither shifts (graph 4)
Both shift	sure where both push; other depends on the bigger shift
Other curve steep	P moves more, Q less. Vertical (planted crops, seats): only P.
Market demand	add buyers' quantities at each P

Own price change: move along. Anything else: shift.

Households and the spending rule

graph 9

- Terms: MU_a = extra satisfaction from one more unit of good a. P_a = price of good a. $MU_a \div P_a$ = satisfaction per dollar on a. Budget line = what your income can buy.

If the question asks	Do this
Rational? (MU, P)	$MU \div P$ each. Equal: yes. Else more of the higher one, less of the other.
P_a rises or falls	up: less a (MU_a rises), more b. Down: more a, less b.
Heatwave, taste	D shifts out: more of it, less of the other
MU from total	change per unit; total peaks at $MU = 0$
Budget line	intercepts income ÷ P (graph 9)
Best transfer	to equal MUs ($MU = 1 \div c$: equal c). Couples, not strangers.

$MU_a \div P_a = MU_b \div P_b$: equal satisfaction per last dollar.

- TRAP** Mistake: more a raises MU_a . Correct: it lowers it.

Elasticity

graph 3

- Terms: $\epsilon = \% \Delta Q \div \% \Delta P$. $\% \Delta = (\text{new} - \text{old}) \div \text{old}$. $|\epsilon| = \epsilon$ without its minus sign. **Elastic:** $|\epsilon| > 1$. **Inelastic:** $|\epsilon| < 1$. **Unit elastic:** $|\epsilon| = 1$. **Spending** = $P \times Q$.

If the question asks	Do this
ϵ from two points	$\% \Delta$ from the starting values. $\epsilon = \% \Delta Q \div \% \Delta P$. Write " $\epsilon = -0.5$, $ \epsilon < 1$: inelastic."
ϵ at a point on a line	$\epsilon = (1 \div \text{slope}) \times (P \div Q)$: elastic high on a line, unit at the midpoint. Vertical: 0. Flat: infinite.
P rises: spending?	Inelastic: rises. Elastic: falls. Small change: $\% \Delta P + \% \Delta Q$. Big: $P \times Q$ before and after.
Given ϵ and $\% \Delta Q$, find $\% \Delta P$	$\% \Delta P = \% \Delta Q \div \epsilon$. $\epsilon = -0.4$, Q down 2%: P up 5%.
Shape of D or S?	Steep: no close substitute, necessity, short run. Flat: substitutes, long run.

Inelastic = steep, elastic = flat.

Surplus, efficiency, DWL

graph 1

- Terms: **WTP** = most a buyer would pay (D = MB = WTP; S = MC). **CS** = $WTP - P$, **PS** = $P - MC$, summed over units. **Efficient** = most CS + PS. **DWL** (deadweight loss) = surplus lost versus the efficient Q.

If the question asks	Do this
Find CS, PS	Solve D = S. CS = $\frac{1}{2} \times Q_1 \times (D \text{ intercept} - P_1)$. PS = $\frac{1}{2} \times Q_1 \times (P_1 - S \text{ intercept})$.
Why is it CS (PS)?	Buyers value units at WTP, pay P. Area under MC = variable cost, so PS = revenue - it.
Is it efficient?	Yes if MB = MC, highest-MB buyers buy, lowest-MC sellers sell. Not always fair.
Shade the DWL	Triangle from actual to efficient Q (too much or too little): $\frac{1}{2} \times Q \text{ gap} \times (MB - MC)$.

- TRAP** Mistake: Q in thousands, area 25 = \$25. Correct: \$25,000.

Price controls

graph 4

- Terms: **Ceiling**, **floor** = legal maximum, minimum price; **binding** if ceiling below P_1 , floor above. Q_S , Q_D = supplied, demanded at it. **Shortage** = $Q_D - Q_S$. **Surplus** (excess supply) = $Q_S - Q_D$. **Misallocation** = wrong buyers or sellers trade.

If the question asks	Do this
Curve shift?	Neither. Horizontal line at the price.
Q traded?	The smaller: Q_S (ceiling), Q_D (floor).
Ceiling: CS, PS, DWL	CS = $\frac{1}{2} \times [(D \text{ intercept} - \text{ceiling}) + (D \text{ at } Q_S - \text{ceiling})] \times Q_S$. PS = $\frac{1}{2} \times Q_S \times (\text{ceiling} - S \text{ intercept})$. DWL = $\frac{1}{2} \times (Q_1 - Q_S) \times (D - S \text{ gap})$.
Do buyers gain?	CS gains a rectangle, loses a triangle: either way. PS falls.
Loss beyond DWL?	Misallocation. Ceiling: queues or luck pick buyers. Floor: not the lowest-MC sellers.
Gouging law; low price with a limit	A ceiling: shortage, the limit rations. Model misses fairness, market power.

Taxes and subsidies

graphs 5, 6

- Terms: **t**, **s** = tax, subsidy per unit. Q_2 = new Q. **P_{buyers}** = price paid. **P_{sellers}** = price kept = $P_{\text{buyers}} - t$ (or + s). **Statutory incidence** = who pays the government. **Incidence** = who really bears it. **Revenue** = $t \times Q_2$.

If the question asks	Do this
Draw it; Q2 and prices	Graph 5. Solve D = S + t. P_{buyers} on D at Q_2 ; $P_{\text{sellers}} = P_{\text{buyers}} - t$.
On buyers or sellers?	Same result. Payroll tax: 50-50 by law, workers bear most.
Who bears more?	Steeper side. Flat S or vertical D: buyers all. Vertical S or flat D: sellers all.
Revenue, DWL	Revenue = $t \times Q_2$. DWL = $\frac{1}{2} \times t \times (Q_1 - Q_2)$: small if a curve is steep, 0 if vertical.
Tax in one city	Sellers can go elsewhere: flat S, buyers pay all.
Price against tax scatter	Slope 1: buyers bear all. 0: sellers. Between: shared. Above 1: costs differ too.
Subsidy	Graph 6 (same if paid to buyers). Solve D + s = S. Cost = $s \times Q_2$. DWL = $\frac{1}{2} \times s \times (Q_2 - Q_1)$.

Steeper side bears more of the tax.

Competitive firms

graph 10

- Terms: **Price taker** = too small to move P, so **MR** (marginal revenue) = P. **Fixed cost** = paid even at zero output. **Variable cost** = area under MC. **Profit** = PS - fixed cost.

If the question asks	Do this
How much to make?	Set $P = MC$, solve for q.
Should it enter?	Yes if $PS > \text{fixed cost}$. Fixed cost never moves q.
Cost up for all firms	Each MC rises more than P: less q, less profit, then exit.
n identical firms	Market $Q = n \times \text{one firm's } q$, at each P.
Long run, free entry	$P = \text{minimum average cost}$, profit 0.
Fixed cost a barrier?	Big for 1 to 2 years, small over 5.

Produce where P = MC. Enter only if PS > fixed cost.

- TRAP** Mistake: profit = PS. Correct: profit = PS - fixed cost.

Monopoly

graph 11

- Terms: **Market power** = can raise P without losing every buyer. **Barrier to entry** = keeps rivals out (patent, unique product, big fixed cost). Demand $P = a - bQ$ ($a = P$ intercept, $b = \text{slope size}$) gives $MR = a - 2bQ$. **Qm**, **Pm** = monopoly Q, P. **Qc** = efficient Q ($D = MC$).

If the question asks	Do this
Why is MR below P?	One more sale cuts P on earlier units.
Find Qm and Pm	MR: same intercept, twice as steep. Qm at $MR = MC$; Pm up on D.
Profit?	$Pm \times Qm - \text{area under MC} - \text{fixed cost}$.
More sold? DWL?	Yes, $Qc > Qm$. DWL: D to MC, Qm to Qc.
Given actual P, Q, MC	It is (Qm, Pm): $a = 2Pm - MC$, $b = (Pm - MC) \div Qm$.
Demand rises	Qm, Pm rise. No supply curve: one point per D.
Per-unit subsidy s	MC falls by s: Q up, P down, DWL down.
Patent? Antitrust?	Patent pays for invention; antitrust targets dominance.

Qm at MR = MC, Pm from D. Flat MC = c: Qm = (a - c) ÷ 2b, Pm = (a + c) ÷ 2.

Externalities

graph 12

- Terms: **Externality** = cost or benefit to outsiders, not in the price. **PMC** = private marginal cost (seller's, S). **EMC** = external marginal cost = harm to others per unit ("damage of \$x"). **SMC** = social marginal cost = $PMC + EMC$. **PMB**, **EMB**, **SMB** = same for benefits ($PMB = D$). **Q*** = efficient Q.

If the question asks	Do this
Q1 versus Q*	Market Q1: D = PMC. Q*: SMB = SMC. Negative: $Q_1 > Q^*$. Positive: $Q_1 < Q^*$.
Optimal pollution 0?	"Usually not: $Q^* (SMC = SMB) > 0$."
DWL	$\frac{1}{2} \times EMC \times (Q_1 - Q^*)$. Positive: $\frac{1}{2} \times EMB \times (Q^* - Q_1)$.
Tax that fixes it	EMC at Q*. Positive: subsidy EMB at Q*.
Total surplus	CS + PS + tax revenue - EMC × Q.
With monopoly	Qm (MR = PMC) versus Q* (D = SMC): equal by luck.
Carbon tax size	Heavy discounting of future harm: small tax.
Coase, cap and trade	Property rights, bargaining. Cap = Q*.

Number problems with straight lines

- Terms: D: $P = a - bQ$. S: $P = c + dQ$. Q1 before, Q2 after.

Task	How
Equilibrium	Demand P = supply P. Solve Q, then P.
CS, PS	$\frac{1}{2} \times Q \times (a - P)$; $\frac{1}{2} \times Q \times (P - c)$. Flat S: PS = 0.
Tax t	Demand P = supply P + t gives Q2. Revenue $t \times Q_2$. DWL $\frac{1}{2} \times t \times (Q_1 - Q_2)$.
Subsidy s	Demand P + s = supply P. Cost $s \times Q_2$. DWL $\frac{1}{2} \times s \times (Q_2 - Q_1)$.
Ceiling	Traded Q_s . Shortage $Q_d - Q_s$. CS = $\frac{1}{2} \times [(a - \text{ceiling}) + (\text{demand P at } Q_s - \text{ceiling})] \times Q_s$.
Monopoly	$a - 2bQ = MC$ gives Qm; Pm from demand. Profit (flat MC): $(Pm - MC) \times Qm$.
Externality	Demand = supply + EMC gives Q*. DWL $\frac{1}{2} \times EMC \times (Q_1 - Q^*)$.
Average P	Revenue ÷ Q in thousands: areas × 1,000.

Check: CS + PS + revenue + DWL = total surplus before.

- $a = 10 - Q$, $P = Q$: $Q = P = 5$, CS = PS = 12.5. Tax 2: $Q = 4$, revenue 8, DWL 1.